

Pharma Drug Sales Performance Dashboard

A Power BI Data Visualization Project

TOOL USED

Microsoft Power BI Desktop

DATA FORMAT

CSV (Daily, Weekly, Monthly, Hourly Sales)

DOMAIN

Pharmaceutical Retail Sales Analytics

DOCUMENT TYPE

Project Documentation

1. Project Overview

This project presents an interactive sales performance dashboard built in Microsoft Power BI for a pharmaceutical retail dataset. The dashboard consolidates hourly, daily, weekly, and monthly sales records across eight drug categories to surface trends, seasonality, and category-level performance for business decision-making.

The goal was to design a single-page executive dashboard that answers common business questions — which drug categories sell best, how sales vary by hour and week, what the monthly and yearly trends look like, and which category drives the most revenue — using only visuals, measures, and interactivity native to Power BI.

Drug Categories (ATC Codes) Covered

Code	Category
M01AB	Anti-inflammatory / anti-rheumatic (acetic acid derivatives)
M01AE	Anti-inflammatory / anti-rheumatic (propionic acid derivatives)
N02BA	Analgesics — salicylic acid derivatives
N02BE	Analgesics — pyrazolones and anilides
N05B	Psycholeptics — anxiolytic drugs
N05C	Psycholeptics — hypnotics and sedatives
R03	Drugs for obstructive airway diseases
R06	Antihistamines for systemic use

2. Data Source

The raw data was provided as four separate CSV extracts, each aggregating the same underlying pharmaceutical sales transactions at a different time granularity:

File	Grain	Rows	Key Columns
salesdaily.csv	Daily	~2,100	datum, 8 drug categories, Year, Month, Hour, Weekday Name
saleshourly.csv	Hourly	~50,500	datum (date+time), 8 drug categories, Year, Month, Hour, Weekday Name
salesweekly.csv	Weekly	~300	datum (week start date), 8 drug categories
salesmonthly.csv	Monthly	~70	datum (month end date), 8 drug categories

Each row records the quantity/value sold per drug category for a given time bucket, along with calendar attributes (Year, Month, Hour, Weekday Name) that support time-intelligence analysis directly in Power BI without extra date tables.

3. Tools & Technology Stack

Category	Tool
BI / Reporting Platform	Microsoft Power BI Desktop
Data Source Format	CSV (comma-separated values)
Data Transformation	Power Query Editor (M language, GUI-driven)
Calculations	DAX (Data Analysis Expressions)
Visualization Types	Stacked column, stacked area, stacked bar, pie, combo line + clustered column, card KPIs, table, slicer

4. Project Workflow

The dashboard was built end-to-end in Power BI Desktop, from raw CSV ingestion to a fully interactive report page. The workflow followed the stages below.

4.1 Data Ingestion & Cleaning

Step 1 — Import Data

All four CSV files (daily, hourly, weekly, monthly sales) were loaded into Power BI using Get Data → Text/CSV, creating four separate tables.

Step 2 — Power Query Transformation

Each table was cleaned individually in Power Query: the date column was reformatted to a proper Date type (it originally loaded as text/incorrect format), blank rows and duplicate rows were removed, columns and tables were renamed for clarity, and rows containing errors were filtered out.

4.2 Report Canvas Setup

Step 3 — Title & Theme

A text box was added for the report title ("Pharma Drug Sales Performance") with custom font colour and background colour, and the overall canvas background colour was set to give the report a consistent look.

4.3 Core Visuals

Step 4 — Weekly Sales by Category (Stacked Column Chart)

Built from the Sales Weekly table. A Week Number measure was created, the stacked column chart was configured with category on the legend, a Top-N filter was applied to show only the first 10 weeks, and gradient colours were applied per category.

Step 5 — Total & Average Monthly Sales (Stacked Area Chart)

Built from the Sales Monthly table with Month on the x-axis. Two DAX measures were created: **Total Monthly Sales** (sum of all drug category sales per month) and **AVG Monthly Sales** (Total Monthly Sales divided by 8, the number of categories), with data markers and value labels enabled for readability.

Step 6 — Quantity Sold Hourly by Category (Stacked Bar Chart)

Built from the Sales Hourly table with Hour on the y-axis and all eight drug categories stacked on the x-axis, with custom fonts and colour formatting applied.

Step 7 — Daily Sales by Category (Pie Chart)

Built from the Sales Daily table, summing sales across all categories and displaying each category's contribution as a percentage of total daily sales.

Step 8 — Yearly Sales Contribution by Category (Stacked Area Chart)

Built from the Sales Monthly table with Year on the x-axis and the sum of each category on the y-axis, showing how category mix shifts year over year.

Step 9 — Quarterly Sales vs Total Sales (Combo Chart)

A combined line-and-clustered-column chart was built from the Sales Monthly table: Quarter on the x-axis, category sales as clustered columns, and the Total Monthly Sales measure plotted as a line to compare quarterly category performance against the overall trend.

Step 10 — Daily / Weekly / Monthly Summary Table

A table visual combining Total Daily Sales, Weekly Sales, and Total Monthly Sales measures side by side, with custom visual styling.

Step 11 — Year Slicer

A slicer on the Year field was added so users can filter the entire report to a specific year (2014–2019).

5. Key DAX Measures

The following DAX measures power the KPI cards and summary visuals on the dashboard:

Total Monthly Sales

Total Monthly Sales = SUM of each drug category's sales, aggregated by month (sum across the 8 category columns in Sales Monthly).

AVG Monthly Sales

AVG Monthly Sales = DIVIDE([Total Monthly Sales], 8, 0)

Weekly Sales Contribution by N02BE

Weekly Sales Contribution By N02BE = DIVIDE('Sales Weekly'[Weekly Sales], SUM('Sales Weekly'[N02BE]), 0) * 100

Total Daily / Weekly Sales measures

Supporting measures aggregating Sales Daily and Sales Weekly totals, used in the summary table and KPI cards.

6. KPI Cards

Four KPI cards give an at-a-glance summary at the top of the dashboard:

AVG MONTHLY SALES	WEEKLY CONTRIB. (N02BE)	TOTAL MONTHLY SALES	PEAK HOURLY QTY (N02BE)
Rs. 15.8K	Rs. 202.5	Rs. 126.5K	251

- **Average Monthly Sales** — mean monthly revenue across all categories.
- **Weekly Sales Contribution by N02BE** — share of weekly sales driven by the N02BE (analgesics) category.
- **Total Monthly Sales** — cumulative monthly revenue across the dataset.
- **Highest Count of Drug Sold on Hourly Basis** — peak hourly quantity sold, again led by N02BE.

7. Final Dashboard

The completed single-page dashboard, titled “**Pharma Drug Sales Performance**,” brings together KPI cards, a year slicer, and seven visuals covering weekly, hourly, daily, monthly, quarterly, and yearly views of sales by drug category.

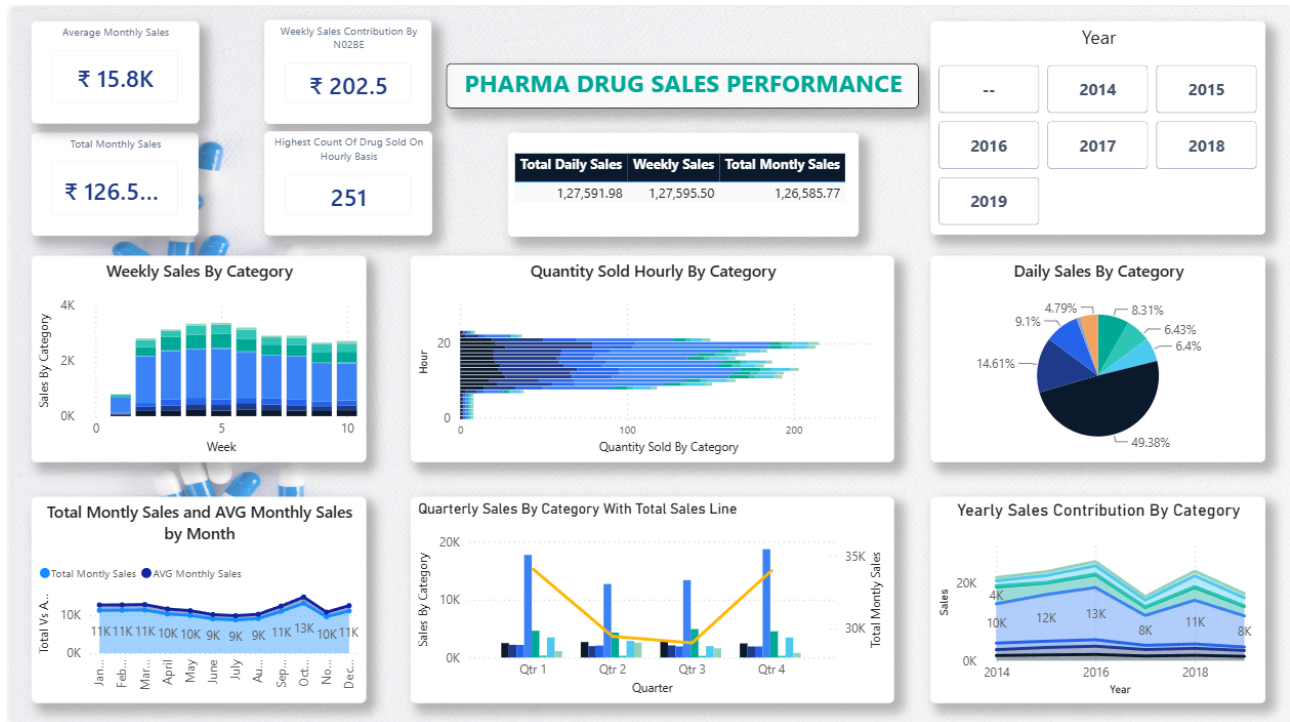


Figure 1: Pharma Drug Sales Performance dashboard (Power BI).

8. Key Insights

- N02BE (analgesics — pyrazolones and anilides) is the dominant category, accounting for roughly half of daily sales share and leading hourly sales volume.
- Monthly sales show clear seasonality, dipping mid-year and recovering toward the end of the year, as seen in the Total vs Average Monthly Sales trend.
- Weekly sales ramp up sharply in the first weeks of the year and then stay in a relatively stable band across the first 10 weeks shown.
- Quarterly performance fluctuates, with Qtr 1 and Qtr 4 generally outperforming Qtr 2 and Qtr 3 on the total sales line.
- Year-over-year category contribution stays broadly consistent from 2014 to 2019, indicating a stable product mix over time.

9. Repository Structure (Suggested)

For the GitHub repository, the following structure is suggested to keep the project organised and easy to navigate:

```
pharma-sales-dashboard/  
|-- data/      (salesdaily.csv, saleshourly.csv, salesweekly.csv, salesmonthly.csv)  
|-- dashboard.pbix  (Power BI report file)  
|-- screenshots/  (dashboard preview images)  
|-- docs/      (this project documentation PDF)  
`-- README.md
```

10. Conclusion

This project demonstrates a complete Power BI workflow — from raw multi-grain CSV data, through Power Query cleaning and DAX measure design, to a polished, interactive dashboard with KPI cards, slicers, and a varied set of visuals. It provides a reusable template for pharmaceutical or retail sales reporting and can be extended with forecasting, additional drill-throughs, or a mobile layout.